

RF-Controlled Aircraft

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Presented here is a remote-controlled aircraft project based on Arduino and 433MHz RF modules controlling a brushless DC motor

and three servo motors. It comprises an Arduino-based remote control at the transmitter's end and an Arduino-based aircraft at the receiver's end. The aim of this project is to develop a 4-channel wireless control system.

Controlling multiple servo motors using XBee RF modules is very sophisticated and

robust but a bit costlier. For cheaper solution, we used a pair of simple 433MHz RF transmitter and receiver modules.

Circuit and working

Fig. 1 shows the circuit diagram of an RF-controlled aircraft's transmitter side and Fig. 2 shows the circuit diagram

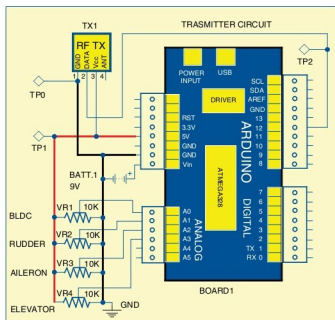


Fig. 1: Circuit of the RF-controlled aircraft (transmitter's side)

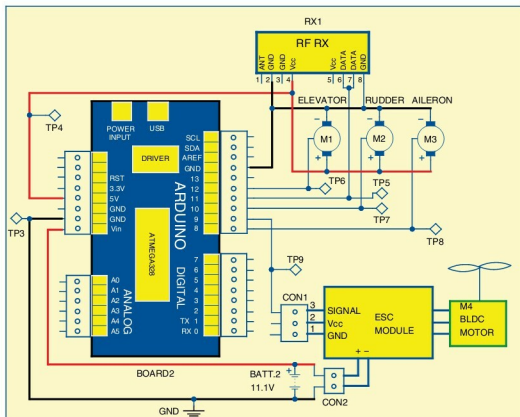


Fig. 2: Circuit of the RF-controlled aircraft (receiver's side)

Test Points

Test point	Details
TP0	0V
TP1	5V (with respect to TP0)
TP2	Transmitted data corresponding to analogue input at A0-A3
TP3	0V
TP4	5V (with respect to TP3)
TP5	Received data
TP6	PWM signals for servo motor (M1)
TP7	PWM signals for servo motor (M2)
TP8	PWM signals for servo motor (M3)
TP9	PWM signals for BLDC motor (M4)

of the receiver side. The circuits are built around Arduino Uno (board1 and board2), a pair of 433MHz RF modules (TX1 and RX1), ESC (electronic speed controller) module, three servo motors (M1-M3), a BLDC motor (M4) and a few other components.

The transmitter's side is driven by a 9V PP3 battery and the receiver's side by an 11.1V LiPo battery, which is used

to power the brushless DC motor (BLDC motor) through the ESC module.

The Arduino board1 receives power supply from the 9V PP3 battery and Arduino board2 from the 11.1V battery at their respective Vin input pins.

There are four potentiometers on the transmitter's side which are used for sending different control signals to the receiver's side through the RF modules. The control signals received by the receiver are processed by the micro-controller in the Arduino, which in turn controls the BLDC motor, the servo motors for rudder, aileron and elevator of the aircraft.

Arduino Uno board. Arduino is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for artists, designers,

